

Advantages of Digital Control in Analogue Power Supplies

Whatever the level of integration, digital control can accelerate the time-to-market by simplifying design changes and providing a platform for a range of applications

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The increased performance of digital control, combined with the lower cost-point of low pin-count, easy-to-use microcontrollers, is challenging the use of analogue for some power supply functions.

Supplying power to a device, board or system is, by definition, an analogue function that is characterised by parameters such as voltage, current and efficiency. Digital control is, however, increasingly being used in the main power-control loop, which raises the question "which other power supply functions could benefit from digital control and which are best left with the proven analogue methods?"

Four levels of digital control

There are four levels of digital integration in power supplies. The most basic, Level I integration, adds simple functions that are difficult to achieve with analogue components. A 6-pin microcontroller, for example, can provide a pulse-width modulation (PWM) waveform that ramps from 0 to 100 per cent duty cycle to provide a soft-start function to the switcher circuitry.

Level II integration provides a secondary management function around the conventional analogue circuit. In this case, the digital controller monitors the output parameters and uses existing external controls to enhance the functionality of the power supply, whilst the power-control loop remains completely analogue. This is accomplished by using standard microcontrollers with an integrated analogue-to-digital converter (ADC) for measuring the various parameters.

Level III integration places the switcher circuitry on-board the microcontroller. By using a more specialised digital controller with integrated switching circuitry and analogue functions, the microcontroller can manage the switching and gain, whilst the implementation of the feedback loop remains primarily analogue.

At the Level IV integration is a complete digital control with all parameters digitised and analysed by the controller to provide the appropriate outputs. This typically requires a DSP with high-speed ADCs and PWM outputs.

The appropriate level of digital integration depends on the specific requirements of the design. Digital controllers excel when making decisions and handling "what if" conditions. When statements such as "if X voltage is higher than Y, then adjust Z" are used to describe the circuit, the best option will probably be to use a microcontroller to perform that function. This secondary monitoring of power-supply parameters and reacting to any exception is one of the most practical uses of microcontrollers in power supplies.

Digital controllers are also useful for sequencing events or timing-related functions. Microcontrollers run from a clock, which makes time measurement and event execution at specified times fairly straightforward. For example, changing the over-current protection level during startup and then throttling it back after 20 ms can be easily achieved with a Level II design that uses a low pin-count microcontroller with an internal oscillator. Again, the digital-control section fits as a complementary peripheral function to the main power supply control loop.

Improved monitoring and time-to-market

Once in the power supply, there are other advantages to be gained by using digital controllers. Monitoring and diagnostics of the supply can be communicated to the rest of the system. Since many of the latest microcontrollers integrate a data EEPROM, event counters and logs can be stored and extracted at a later stage to provide valuable information about failures and power-supply usage. The EEPROM is also valuable for storing calibration values that can be used to linearise a temperature coefficient or reduce the cost of a voltage reference.

Possibly, however, the most useful advantage of using digital control for power supplies is found during the design stage. The ability to change behaviours simply by reprogramming the controller accelerates the rate of change and, eventually, time-to-market. One design can be used for a range of applications by simply changing the firmware, with no need for hardware or component changes. This results in lower inventory costs and fewer manufacturing setup changes. It also replaces manual calibration with electronic calibration, which, in turn, speeds customisation of a design, providing substantial additional cost savings.

Thus advantages of digital control in analogue power supplies range from simplifying difficult analogue functions such as PWM for soft-start to providing secondary management functions or complete digital control of all parameters. ●

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